[Feature Name]

Author: Person

Contributors: Person, Person

Status: Brainstorming

Version History

Version	Changes	Author	Date
0.1.0	Initial Version	Person	Date

Problem Statement

1.Problem

Da Nang is already operating at scale as a destination: in the first seven months of 2025 alone, the city welcomed **10.7 million visitors, including over 4.2 million international arrivals**, generating about **VND 33.8 trillion** in tourism-related revenue.link The city is actively pushing smart tourism and the year-end stimulus programme **"New Da Nang – -New Experience"** to diversify products and apply digital solutions across the service chain.link

However, most "deep" experiences still cluster around a few familiar corridors (beachfront, Ba Na Hills, Hội An), and current digital tools (portals, VR apps, chatbots) are largely informational or per-activity, not a continuous, city-level companion.link This makes a smart AI local buddy naturally **less relevant for fixed-itinerary package tour groups**, but highly valuable for **free independent travellers (FITs) and business visitors**, who want flexible, context-rich guidance on where to stay, work, eat and explore across different districts rather than just following a standard tour script.

---

2. Impact on the City (Public, Private, Governance)

From the public side, Da Nang's tourism programmes explicitly aim to welcome about **17.3 million overnight visitors in 2025**, including **7.6 million international guests**, by diversifying experiences and spreading benefits beyond traditional hotspots.data Without a personalised, city-integrated AI companion for FITs and business travellers, it is hard to steer demand toward new areas, lengthen stays, or systematically support people who want to test Da Nang as a long-term base for work and life.

On the private side, smaller local businesses, cafés, co-working spaces and services outside the main tourist strips remain largely invisible, even though smart-tourism research notes that Da Nang has already built substantial ICT and data infrastructure and is treating smart tourism as a key pillar of its smart-city strategy.quote

From a governance perspective, a city-level AI local buddy gives authorities a new, **privacy-respecting data layer**: aggregated insights into *why* people come (short city break, MICE, scouting a place to live), *where* they want to go, and *what* they struggle with. This kind of data, in line with broader smart-tourism and digital-governance practice, can help Da Nang:

+ Target public services and information more precisely,
+ Identify overcrowding or low-quality services early, and
+ Reduce the risk of negative phenomena (unbalanced development, unmanaged flows, poor visitor experiences) as volumes keep rising. ResearchGate+1

---

3. Why Now

In June 2025, Viet Nam's National Assembly approved a resolution to establish **International Financial Centres (IFCs) in Ho Chi Minh City and Da Nang**, with the framework taking effect on **1 September 2025**.link The IFC is intended to attract international capital and skilled professionals and to reshape Viet Nam's financial role in the region—Da Nang's identity is therefore shifting from "only" a beach destination to a combined **tourism + business + lifestyle hub**. At the same time, the city is highlighted as a pioneer in digital tourism, rolling out a shared big-data system and AI-enabled services to connect authorities and businesses. Thông tấn xã Việt Nam

Globally, AI-powered travel assistants and hyper-personalised, data-driven tourism are becoming mainstream, but **city-branded, city-integrated AI "local buddies" tailored to FITs and business visitors are still rare**.link Launching a Da Nang–specific AI agent now would:

+ Give independent and business travellers a high-value companion that extends their experience beyond classic tour-group corridors,

+ Open up new areas and services across the city for both visitors and long-stay residents

+ Provide exactly the experiential and data foundation Da Nang needs to function not just as a resort city, but as a credible, well-governed **international financial centre**.

Objective

Build a Da Nang–specific AI local buddy that makes the city's smart-city and IFC vision tangible in everyday life, while spreading benefits more evenly across people and places.

1. **Make smart city feel real, not abstract**
   Turn Da Nang's digital-government and smart-city initiatives into something residents and visitors can actually feel: clear guidance on where to go, what to do, which services to use, instead of "smart city" staying as a policy slogan.

2. **Support Da Nang as a work and finance hub, not just a beach stop**
   Give independent travellers and people coming for work (experts, corporate staff, remote workers) a soft landing where culture, lifestyle and daily logistics are easy to understand, aligning with the national plan to build an International Financial Centre in Da Nang.

3. **Spread income and opportunity across the whole city**
   Direct demand beyond a few classic tourist pockets into community-based, eco-tourism and emerging urban areas, so more households and small businesses can participate in tourism and service growth.

4. **Reinforce Da Nang's identity as an international city**
   Increase meaningful contact between locals and a diverse mix of visitors and long-stay residents, building on Da Nang's positioning as a digital, innovative, globally connected city rather than only a seasonal holiday destination.

Key User Scenarios

1. Independent Traveler – Short City Break (3 days in Da Nang)

-Context

A solo traveler from Korea arrives in Da Nang for a 3-day trip. They speak basic English, almost no Vietnamese, and prefer to explore on their own rather than join a tour. They want to move around easily, avoid "tourist traps," and understand basic local info (weather, traffic, opening hours).

-Ideal Journey

+ User scans a QR code at the airport and opens the **web-based AI buddy** (no download, just browser).
+ They see a single, simple interface: **chat box + voice button** with multi-language support.
+ User taps the voice button (in Korean or English): "I'm at the airport, I want to go to '123 Nguyễn Văn Thoại'. Can you help me book a ride?"
+ The AI recognizes the Vietnamese address, confirms the location on a map, and generates a **direct deep-link to Grab** so the user can book with one tap, without typing Vietnamese.
+ On the way, the user continues chatting: "What's the weather like this afternoon? Is there any risk of heavy rain near the beach?" The AI returns a concise answer and suggests the best time window to go out.

+ In the evening, the user asks: “Is the road to Sơn Trà busy now? Any safer route for a motorbike?” The AI responds with up-to-date road guidance and a suggestion to avoid certain hours or areas.

-Benefit

The traveler never feels “stuck” because of language or local logistics. They can move around freely with just voice/chat and a browser, which directly supports Da Nang’s goal of making smart-city services visible and useful to everyday visitors.

---

2. Business Visitor – One Week Work Trip (IFC / Corporate Meetings)

-Context

A regional manager from a multinational company flies to Da Nang for a one-week work trip. They have a tight meeting schedule, want to stay near suitable cafés/co-working spaces, and need quick answers about basic administrative processes (e.g., business registration, tax office location) to report back to HQ.

-Ideal Journey

+ On arrival, the user opens the AI buddy on their laptop (web, no install) from the hotel Wi-Fi. They start with text chat in English: “I’ll be in Da Nang for 7 days for meetings. Which area should I stay in if I want quiet cafés to work and easy access to the city centre?”
+ The AI suggests a few neighbourhoods, with high-level descriptions (business-friendly, walkable, near offices) and links out to map locations.
+ Next, the user asks: “Tomorrow morning I need to visit the tax office and then a co-working space nearby. Can you give me directions and a Grab link?”
+ The AI returns:

- The correct office address in Vietnamese,
- A clear description in English,
- A **Grab deep-link** so they can book a ride without retyping the address.

+ Between meetings, the user asks: "What are the basic steps for a foreign company to open a representative office here?" The AI provides a concise **high-level checklist** and points to the relevant official websites/offices, framed clearly as informational (not legal advice).

-Benefit

The business visitor experiences Da Nang not just as a holiday spot but as an easy place to work and navigate bureaucracy at a basic level. This reduces cultural and practical friction for future investors and professionals, supporting Da Nang's positioning as a regional financial and business hub.

---

3. Long-Stay / Aspiring Resident – Exploring Life in Da Nang

-Context

A remote worker from Europe has already visited Da Nang once as a tourist. Now they return for a month to test if they want to live there longer-term. They need to understand everyday life: areas to live, commute times, basic admin (temporary residence, visa extensions), and how convenient the city feels.

-Ideal Journey

+ User opens the AI buddy from their phone browser on day 1 in Da Nang.
+ Using voice in English, they say: "I'm staying near Hải Châu now. Which neighbourhoods should I explore if I want a quieter area with cafés, gyms and not too far from the beach?"

+ The AI suggests several neighbourhoods with short descriptions (vibe, typical residents, distance to beach/centre) and proposes a **day route**:
    - Morning: cafés in area A,
    - Afternoon: walk/ride around area B,
    - Evening: dinner options in area C.
+ During the day, they ask: “How bad is the traffic if I commute from here to the city centre at 8:30 a.m.?” The AI provides an estimate and suggests better time windows.
+ Later in the week, they ask: “What basic administrative steps do I need if I want to stay here longer as a remote worker?” The AI returns a **high-level overview**: where to find information about visas, temporary residence, and which official channels to follow, with links to the relevant government sites.

-Benefit

The long-stay user gets a realistic, low-friction sense of what it’s like to live in Da Nang day to day—moving around, working, and dealing with basic admin. This makes Da Nang feel like a viable place to live and work, not just a holiday stop, and encourages more balanced, long-term engagement with different parts of the city.

---

4. Local Resident – Everyday Questions and City Services

-Context

A young local resident hears about the AI buddy from a city campaign and wants to use it for quick, everyday tasks: checking weather and road conditions before commuting, and clarifying simple administrative questions (e.g., where to renew a driver’s license or pay certain fees).

-Ideal Journey

+ The resident opens the AI buddy on their phone browser on the way to work.
+ They type: “Is there any heavy rain expected in the next 2 hours near Liên Chiểu? I’m going by motorbike.” The AI gives a short forecast and suggests the safest time window to depart.
+ Later, they ask: “Where do I go to renew my driver’s license, and what documents do I need?”. The AI responds with:
    - The correct office name and address in Vietnamese,
    - A simple checklist of required documents,
    - A **Grab deep-link** for that address if they want to book a ride.

-Benefit

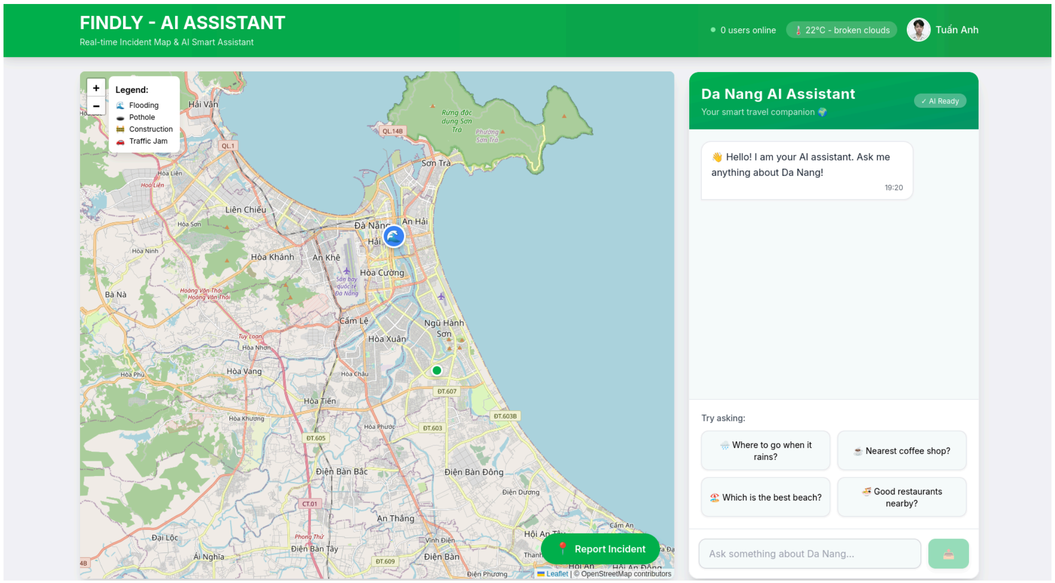
Local residents experience the AI buddy as a simple, low-friction front door to both city information and daily logistics. This reinforces trust in smart-city services and shows that “smart city” is not just for tourists, but for citizens’ everyday life.

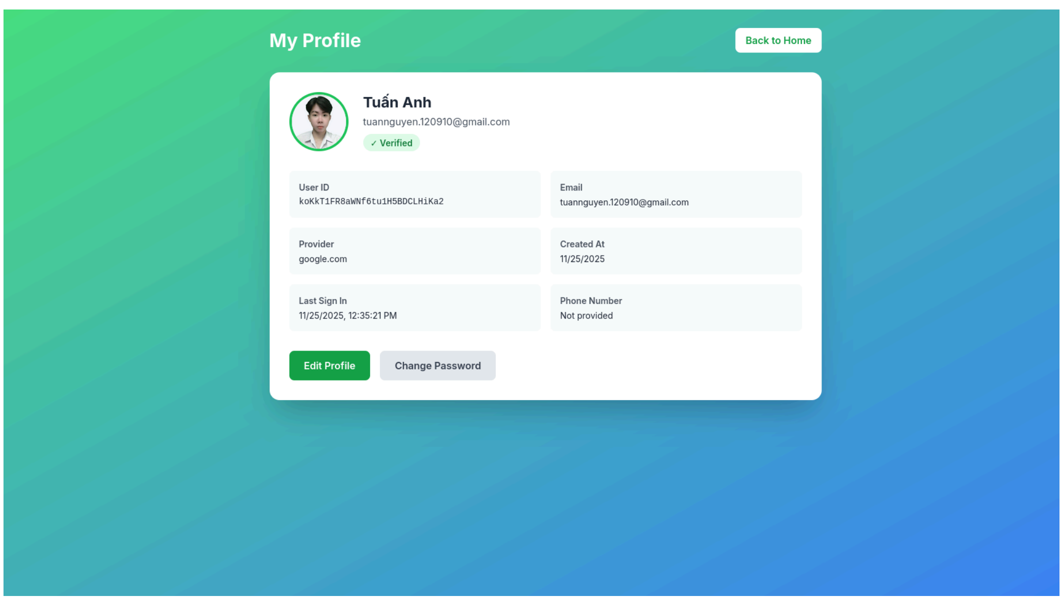
Mockups & Design (optional)

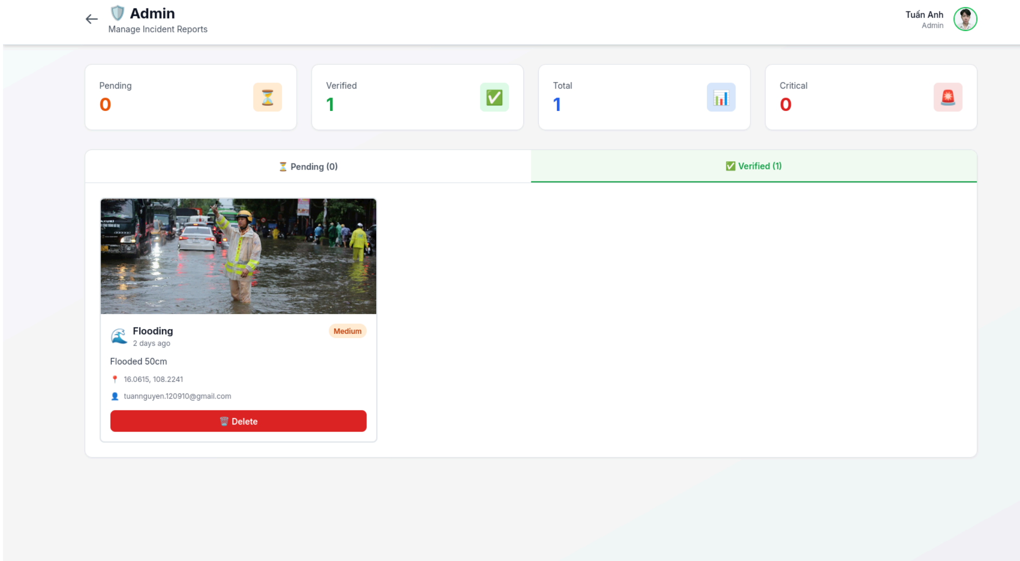
- Logo:

-Product interface:

+ Website

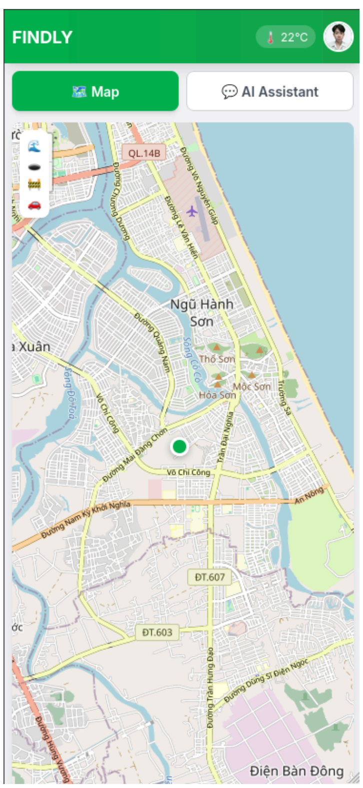




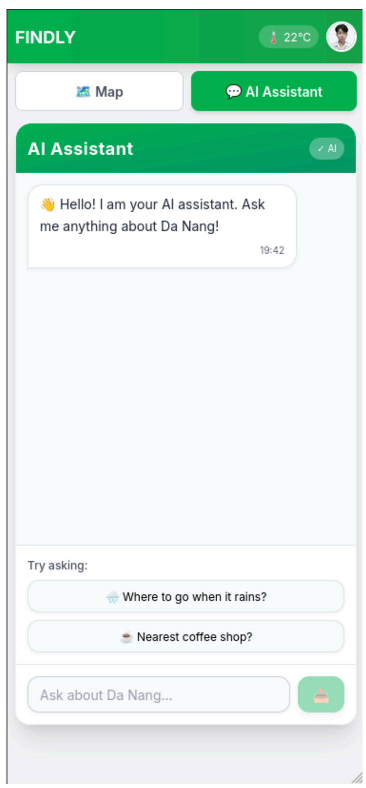


Admin
Manage Incident Reports
Tuấn Anh
Admin
Pending
0
Verified
1
Total
1
Critical
0
Pending (0)
Verified (1)
Flooding
Medium
2 days ago
Flooded 50cm
16.0615, 108.2241
tuannguyen.120910@gmail.com
Delete

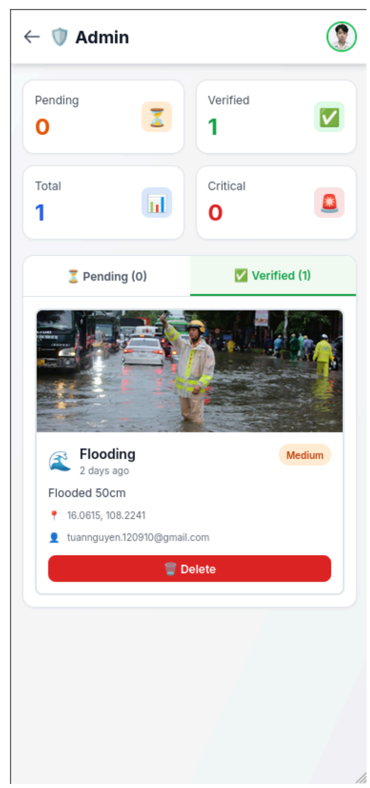
+ **Responsive:**



FINDLY
22°C
Map
AI Assistant
QL.14B
Ngũ Hành Sơn
Xuân
Võ Chí Công
ĐT.607
ĐT.603
Điện Bàn Đông



FINDLY
22°C
Map
AI Assistant
AI Assistant
✓ AI
Hello! I am your AI assistant. Ask me anything about Da Nang!
19:42
Try asking:
Where to go when it rains?
Nearest coffee shop?
Ask about Da Nang...



Admin
Pending
0
Verified
1
Total
1
Critical
0
Pending (0)
Verified (1)
Flooding
Medium
2 days ago
Flooded 50cm
16.0615, 108.2241
tuannguyen.120910@gmail.com
Delete

Solution Architecture

1. Technology Stack:

Component	Technology	Justification
Frontend	Next.js 14 (React, TypeScript) + Tailwind CSS + Leaflet + Socket.IO Client	Next.js provides a robust, scalable foundation with SSR/SSG. TypeScript ensures type safety. Leaflet is an efficient map library for map rendering, and Tailwind CSS enables rapid, responsive styling.
Backend	Node.js (Express) + Socket.IO	Node.js allows a unified language stack (JavaScript/TypeScript) across the application. Express provides a flexible REST API server. Socket.IO ensures real-time communication (WebSockets) essential for immediate hazard reporting updates.

Database	Firebase.	Firebase provides a fully managed, scalable NoSQL database (Firestore/Realtime Database) that is ideal for real-time updates (essential for hazard reporting). It integrates seamlessly with **Cloud Storage** for secure, scalable storage of media assets like user-submitted report photos, offering a unified, serverless backend environment.
Infrastructure	Vercel	Vercel provides seamless integration with Next.js, offering high-performance hosting, automatic Serverless Function scaling, and global CDN for low-latency delivery, which is critical for a fast user experience.
Monitoring		

2. High-Level Architecture

-The system is designed with a microservices-based architecture to ensure scalability, resilience, and independent deployment of core functionalities, primarily centered around the AI Recommendation Engine.

+ **Web Application (Frontend):** This is the user-facing Single Page Application (SPA) built using Next.js/React. It is responsible for user authentication, collecting natural language input, managing component state, displaying interactive map views (via Leaflet), showing real-time hazard overlays, and initiating deep-link calls to the Grab application via web redirection. It utilizes the **Socket.IO Client** for real-time updates and notifications pushed from the backend.
+ **API Gateway:** Acts as the single entry point for all client requests. Its core functions include JWT authentication for securing all services, rate limiting to prevent abuse, and load balancing/routing requests to the correct internal microservice (e.g., /recommendation, /reporting, /profile).
+ **Recommendation Service (AI Agent):** This is the heart of the application
    - **Flow**: It receives a natural language query, normalizes it, and sends it to the Gemini API for contextual interpretation and processing (with system context on Da Nang POIs).
    - **Data Retrieval:** It simultaneously queries the **POI Database** (PostgreSQL/Vector DB) to retrieve a set of highly relevant location candidates based on semantic similarity.
    - **Filtering & Ranking:** The service applies filters (e.g., operating hours, current user location, reported hazards) and uses a custom ranking algorithm (or a second small LLM call) to generate the final, personalized list of 3-5 locations.
+ **Reporting Service:** Manages the lifecycle of user-submitted road reports.
    - **Collection:** Accepts GeoPoint data, type (pothole/construction), and optional image uploads (stored in Supabase Storage).
    - **Validation:** Reports are initially marked new. In the short term, validation uses simple clustering or admin review. For the next phase, we will integrate a dedicated Visual Recognition Model (VRC) to automatically analyze the uploaded image and GPS data to classify the hazard type and veracity, changing the status to verified.

- **Real-Time Push:** Uses Socket.IO to instantly broadcast newly verified hazard locations to the Web App, ensuring all active users see real-time map updates without manual refresh.

+ **Third-Party Integration Services:** Dedicated functions handle external communications.
    - **Grab API Integrator:** Exclusively focuses on constructing the platform-agnostic deep-link URL based on the target destination's coordinates and name. This service ensures the URL format is current and adheres to Grab's specifications for smooth redirection.
    - **Google Maps Integrator:** Used by the Web App for base map tiles (via Leaflet), but this backend service is reserved for complex calculations, such as batch distance/time matrix lookups when generating a multi-stop itinerary or checking road conditions.

Implementation & Feature Breakdown

-This section focuses on the three most critical and complex features: AI Recommendation, Grab Integration, and the User Reporting System.

Key Feature / Component	Description	Technical Details (Simplified)
Feature 1: AI-Powered Recommendation	Allows the user to ask for travel suggestions using natural language (e.g., "I want a quiet coffee shop near My Khe Beach with a view"). The AI processes the query, filters POIs (Points of Interest) by location, mood, and time, and presents a ranked list.	**- Input:** User Query (Text). - **Processing:** Call to the custom Gemini API, utilizing detailed, custom system prompts to generate the recommendation text according to predefined rules (acting as a Da Nang expert agent) + retrieval of structured POI data from the Firebase Database (e.g., using vector embeddings for semantic similarity search). - **Output:** JSON list of 3-5 recommended locations displayed via React components.

Feature 2: Grab Deep Link Integration	Once a location is selected, a single button click generates a direct link that opens the Grab app or redirects to the booking page/app store, ensuring seamless transportation booking.	**Client-Side Action:** The web app constructs a URL (e.g., a universal deep link or a web URL for Grab's booking interface) using the selected location's coordinates and address. **Fallback:** The redirection logic includes error handling to inform the user if the deep link fails to open the application.
Feature 3: Real-Time Road Reporting	Allows users to submit reports on road hazards (potholes, construction) with a photo and location. These reports are shown to other users on the map to help them avoid problem areas instantly.	**Data Model:** Store reports in Firebase with fields: reporter_id, type (pothole/construction),status (new / verified /fixed), timestamp, location (GeoPoint), photo_url. **Real-Time:** The Reporting Service uses Socket.IO to push new verified hazard locations to all connected Web Apps, and Leaflet renders these as interactive markers.

Appendix